

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		To ensure that the cake case has reached terminal [top / maximum....] speed / has stopped accelerating	1			1		1
		(ii)		To check repeatability / identify anomalies (1) [Not: to ensure there are no anomalies] To improve the mean / give a more accurate mean / to reduce the effect of random errors / to allow anomalies to be removed before calculating the mean (1) Reference to 'reliability' → no credit	2			2		2
	(b)	(i)		Substitution into $s = \frac{d}{t}$ i.e. $2.88 = \frac{1.50}{t}$ (1) $t = 0.52$ [s] (1) [accept 0.5, 0.521]	1	1		2	2	2
		(ii)		Appropriate scales on both axes 0.5 g and 0.5 m/s per 2 cm square (1) 5 points plotted correctly (ignore (0,0) to within <1 small square division (2) 4 points plotted correctly (ignore (0,0) to within <1 small square division (1) 3 points plotted correctly (ignore (0,0) to within <1 small square division (0) Smooth curve of best fit to (0,0) < 1 small square division (1)	1	1 1 1		4	4	4
		(iii)		Correct pair of values of speed taken from graph or table for 1.0 g and 2.0 g or 1.5 g and 3.0 g (1) Calculation of ratio for at least 1 other pair of masses e.g. $\sim 2.8 / 1.60 = \sim 1.75$ or $2.94 / 2.54 = 1.16$ (1) Conclusion [must include a valid comparison of ratio to 1.4] – so not always true (1) Alternative route Correct pair of values of speed taken from graph or table for 1.0 g and 2.0 g or 1.5 g and 3.0 g (1) [can be implied by method] Calculation of expected speed, i.e. $2.54 \times 1.4 = 3.56$ or $2.24 \times 1.4 = 3.14$ (1) Valid comparison of speeds, e.g. $3.56 \neq 2.94$, or $3.14 \neq 2.8$ so not always true (1)			3	3	2	3

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	(c)			Valid improvement and justification e.g. Drop the cake cases from a greater height (1) So that the random errors due to timing are a smaller fraction of the value measured (1) Alternative: Use light gates / record with a camera or phone (1) to give a more accurate time measurement / reduce reaction time errors (1) Alternative: Measure the mass of the stack / each case (1) to give a more <u>accurate</u> mass / mass closer to the true value(1) Not: extra data collected			2	2		2
	(d)			Use of weight = mg (1) [allow even if mass not converted] air resistance = 0.025 [N] [on answer line] (1) cao air resistance = weight (1)	1	1 1		3	3	
				Question 2 total	6	6	5	17	11	14